

Fluid Mechanics Course

External flows

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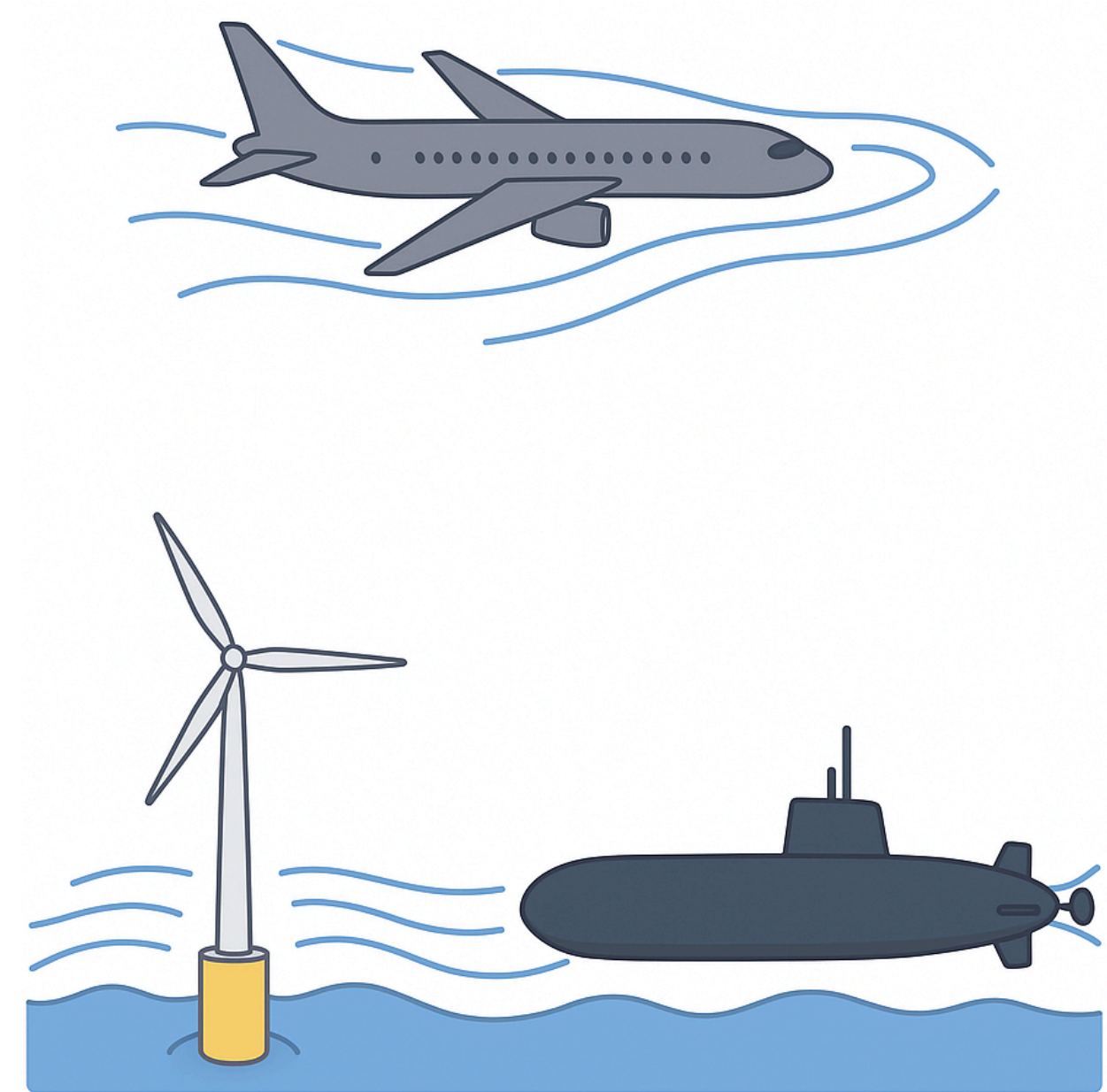
Slides by Jacob Andersen (AAU BUILD) and Jakob Hærvig (AAU ENERGY)

Introduction to External Flows

External flows: Flow around bodies immersed in fluid

Examples:

- Air flow around (aerodynamics):
 - buildings
 - wind turbine blade
 - airplanes
 - cars
- Water flow around (hydrodynamics):
 - offshore wind turbine foundation
 - vessels
 - submarines
 - water turbine



Analysis of External Flows

Accurate description of *external fluid forces*: Key in optimizing designs with regards to, e.g.,

- Power generation (wind, hydro, wave)
- Fuel reductions (airplanes, cars, vessels)
- Structural strength (material usage)

Approaches

- Limited *analytical* techniques
- *Experimental* methods: Wind tunnels, towing tanks, wave bassins etc.
- *Numerical* methods (CFD): Large variation in fidelity

Body Classification and Flow Characteristics

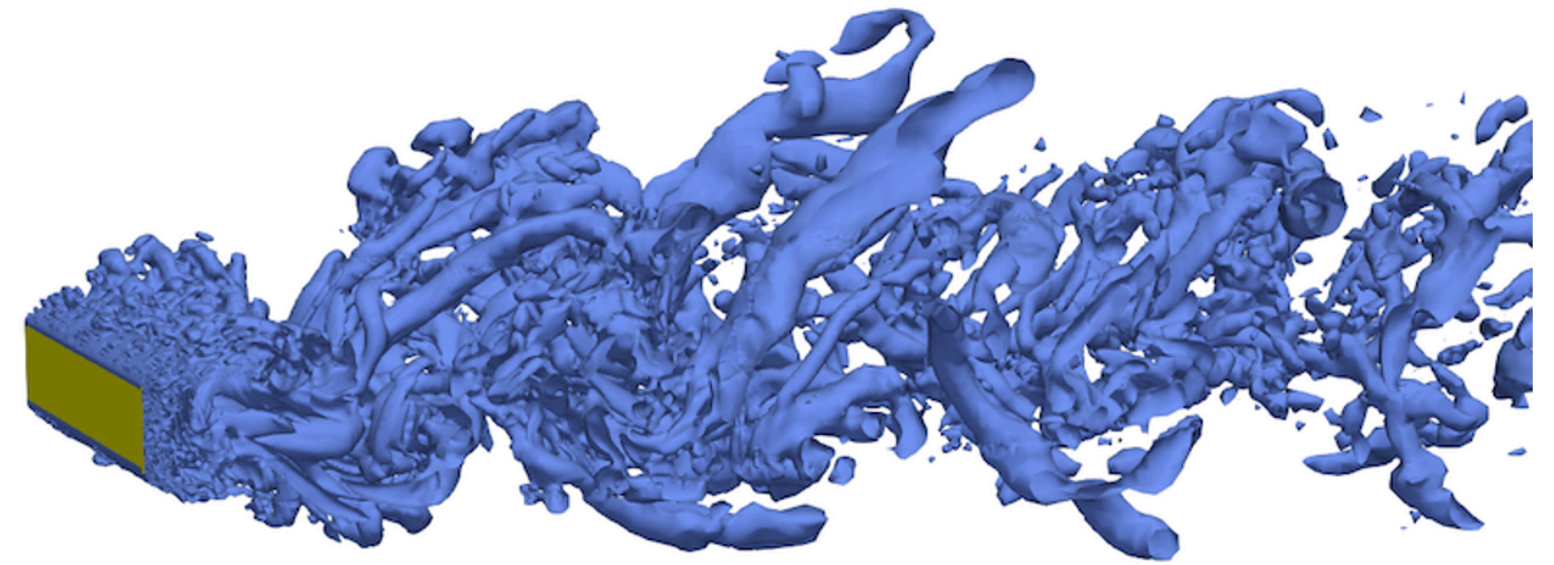
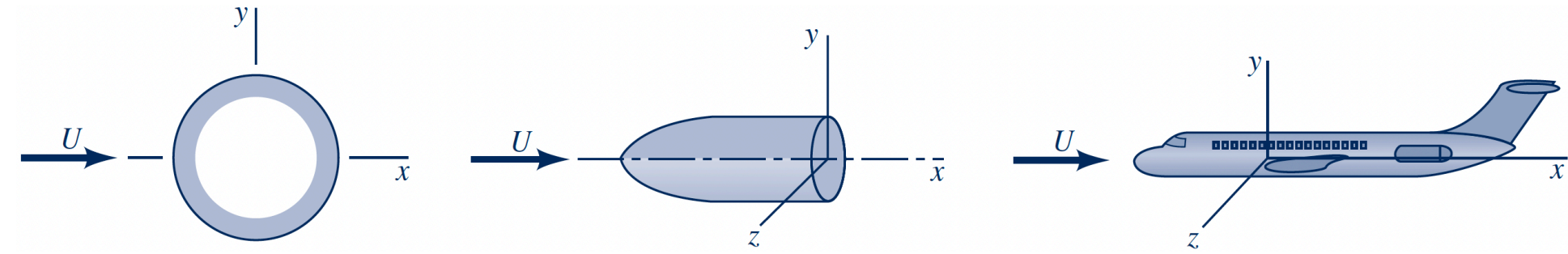
Categorization of bodies:

- *2-D* bodies (infinitely long, constant cross-section)
- *Axisymmetric* bodies (cross-section rotated about symmetry axis)
- *3-D* bodies

Note: Nominal 2-D bodies may need to be modelled as 3-D depending on the turbulence characteristics of the flow

Further categorization:

- *Streamlined* bodies (airfoils, racing cars) - little fluid disturbance
- *Blunt/bluff* bodies (stalled airfoils, buildings, parachutes) - significant fluid disturbance
- Generally less fluid resistance to streamlined bodies



Lift and Drag Concepts

External force on "stationary" body in flow with *upstream velocity* U

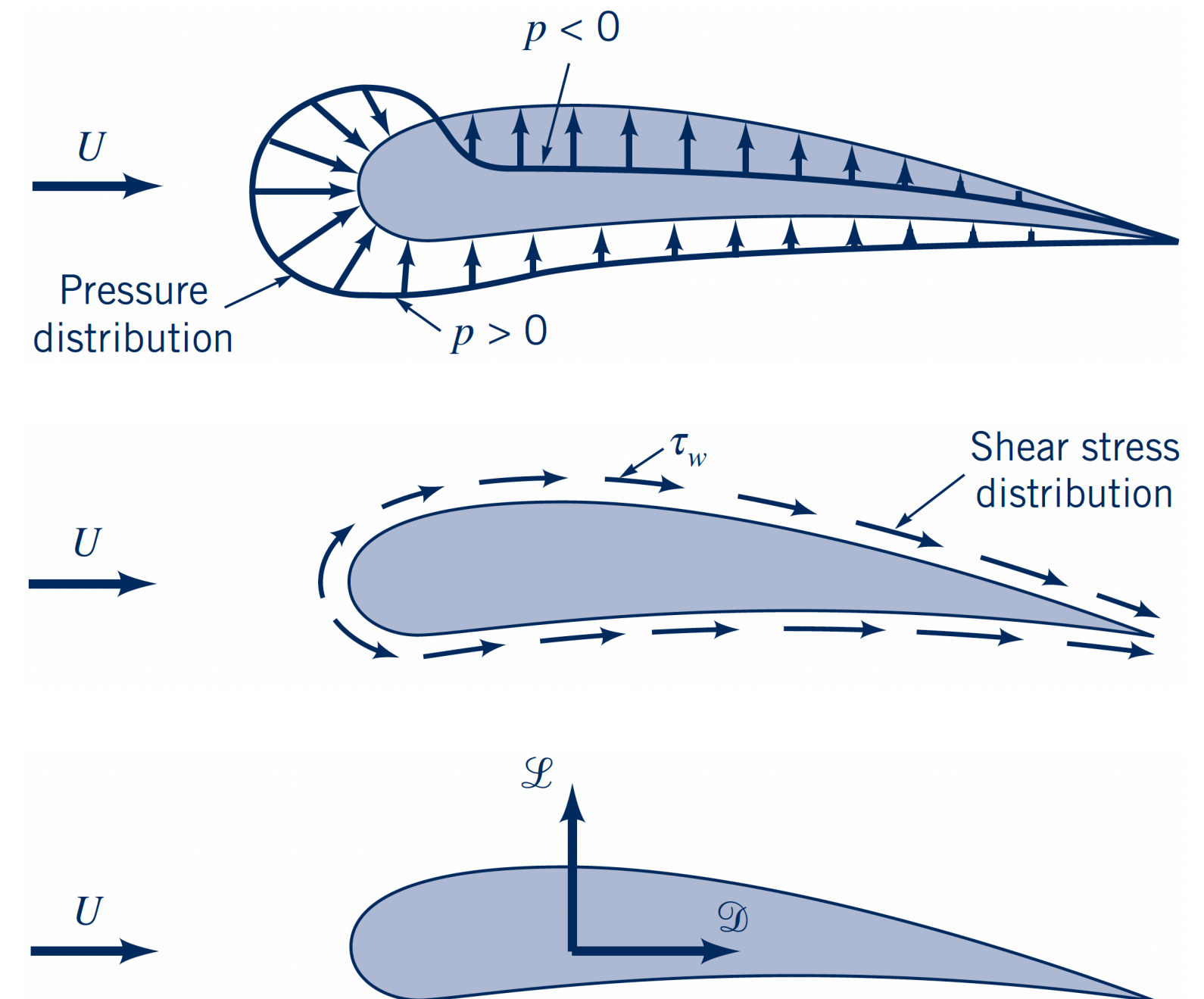
- Normal stresses due to **pressure**
- Wall shear stresses due to **viscous effects**

Pressure and wall shear stress distributions: Useful for detailed design and analysis

Usually we are mostly interested in the *integrated* stresses on the body, i.e., the *resultant force*

- **Drag** $\mathcal{D}$: Component of the resultant force in the direction of the free-stream velocity U
- **Lift** $\mathcal{L}$: Component of the resultant force normal to the free-stream velocity U

p and τ_w vary in magnitude and direction: Contribute to both $\mathcal{D}$ and $\mathcal{L}$



Lift and Drag Coefficients

p and τ_w distributions are *difficult* to obtain (virtually only from *CFD*)

Often, we use *dimensionless coefficients* to estimate drag and lift forces:

- **Drag coefficient:**

$$C_D = \frac{\mathcal{D}}{\frac{1}{2}\rho U^2 A}$$

- **Lift coefficient:**

$$C_L = \frac{\mathcal{L}}{\frac{1}{2}\rho U^2 A}$$

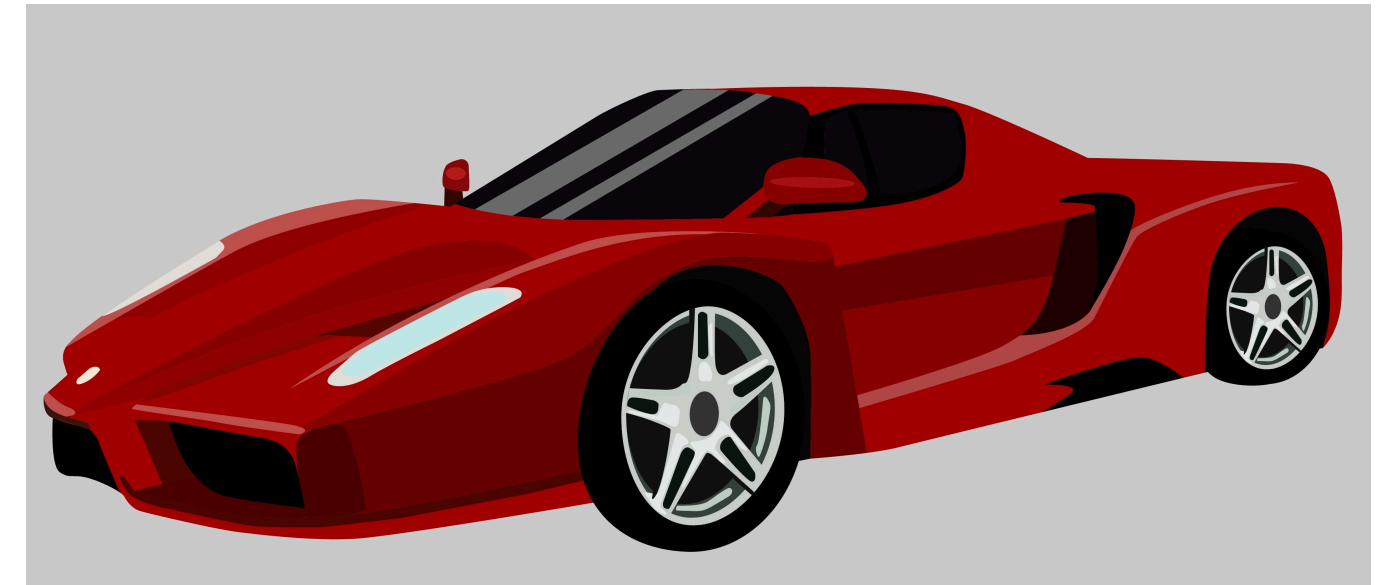
Characteristic area A :

- Frontal area (projected in streamwise direction)
- Planform area (projected normal to streamwise direction)
- Other definitions (wetted surface, etc.)

Power consumption:

$$P = \mathcal{D} \cdot U = \frac{1}{2}\rho U^3 C_D A$$

Scales with U^3 !



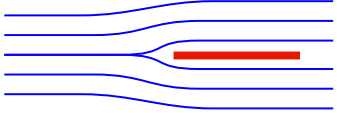
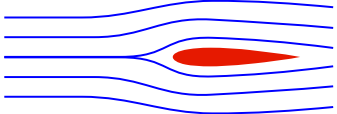
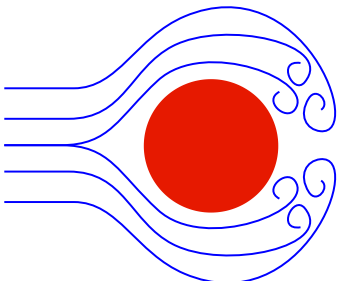
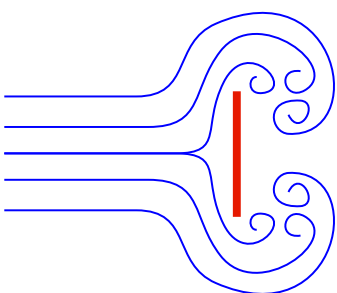
Power required to overcome drag increases rapidly with speed!

Pressure and Friction Drag

Drag can be separated into two components:

- Drag caused by *viscous shear* stresses: **friction drag**
- Drag caused by *pressure* differences: **pressure drag** or **form drag**

Pressure drag dominates for blunt bodies, while friction drag dominates for streamlined

Shape and Flow	Pressure (form) drag	Friction drag
	≈0%	≈100%
	≈10%	≈90%
	≈90%	≈10%
	≈100%	≈0%

Drag (physics)

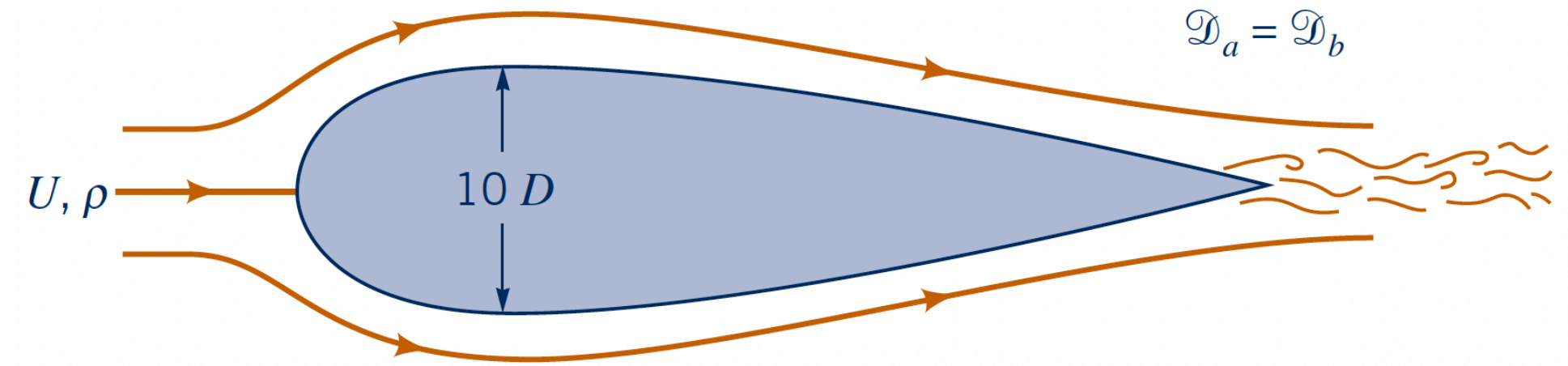
Shape Dependence on Drag

Two bodies where *characteristic lengths* vary with an *order of magnitude*: *Equal drag*!

- Underlines the impact of streamlining
- **Separation causes the drag!**



(a)



(b)

Dimensionless Numbers in External Flows

Shape of body is important, but not the full story.

- **Simple shapes $\neq$ simple flows**
- Simplicity (or lack of) is *governed* by the *flow characteristics*
 - Flow characteristics represented by *dimensionless numbers*:
 - *Reynolds number* (Re): inertia / viscous
 - *Froude number* (Fr): inertia / gravity
 - *Mach number* (Ma): inertia / elasticity (compressibility)
 - *Weber number* (We): inertia / surface tension
- Common orders of magnitude for characteristic lengths and upstream velocities in external engineering flows yields:
 - $10 < Re < 10^9$
 - Rule of thumb: $Re > 100 \rightarrow$ flow dominated by inertial effects

Re -dependency for streamlined body

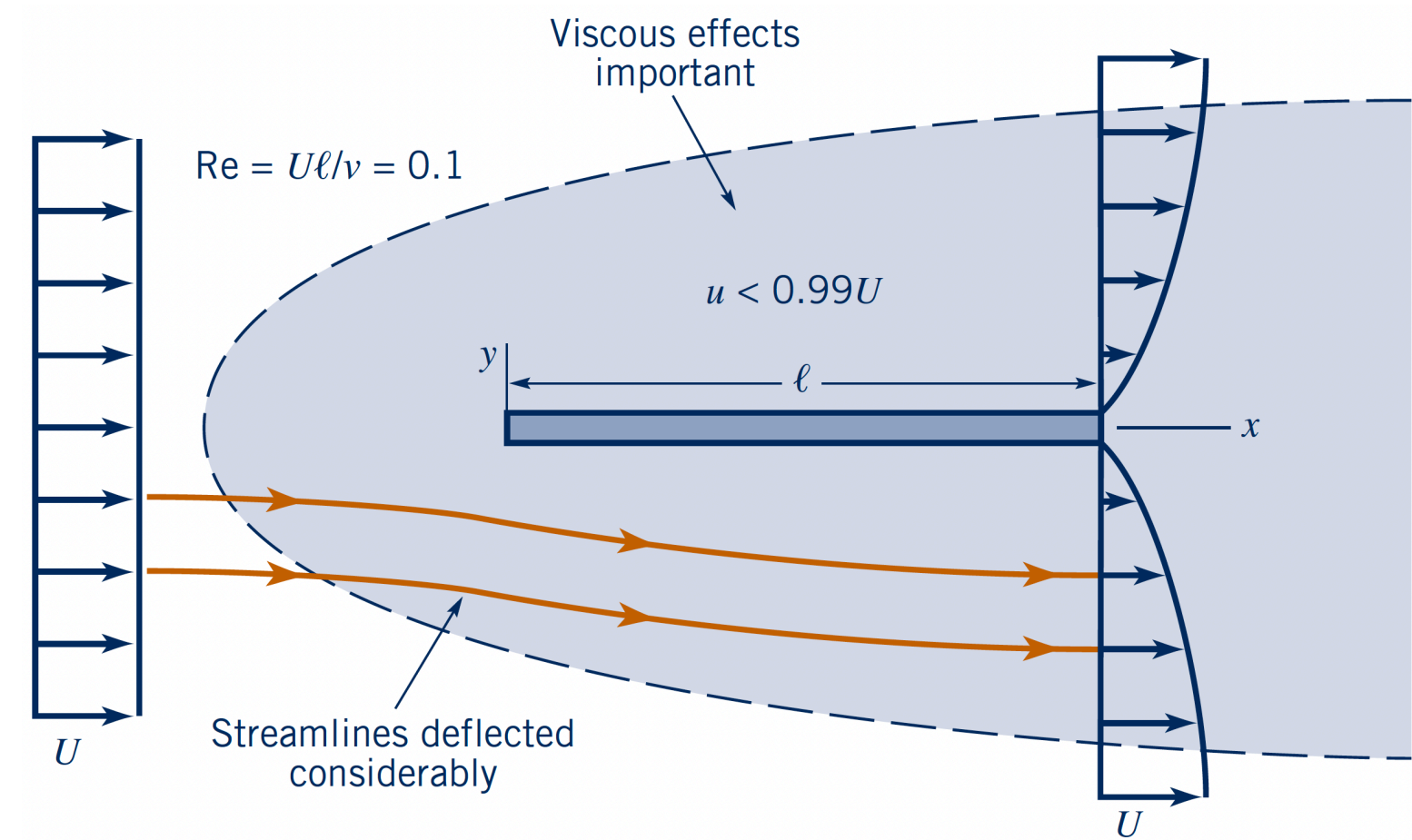
Parallel flow to a flat plate at three different $Re = \{0.1, 10, 10^7\}$.

- Plate length: l
- Upstream velocity: U
- Boundary layer: $u < 0.99U$

Re -dependency for streamlined body

Parallel flow to a flat plate at three different $Re = \{0.1, 10, 10^7\}$.

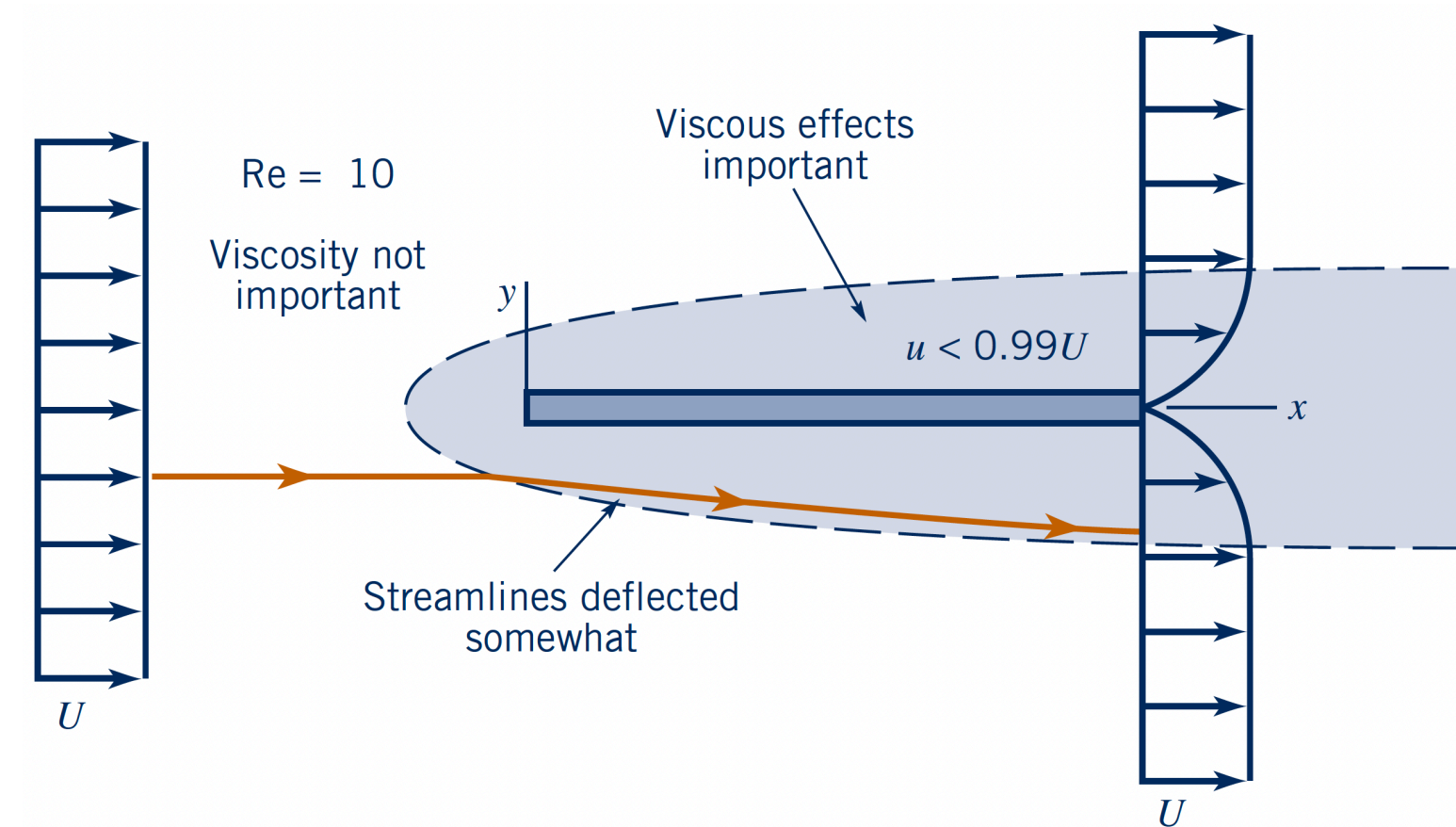
- Plate length: l
- Upstream velocity: U
- Boundary layer: $u < 0.99U$
- Large disturbance of flow



Re -dependency for streamlined body

Parallel flow to a flat plate at three different $Re = \{0.1, 10, 10^7\}$.

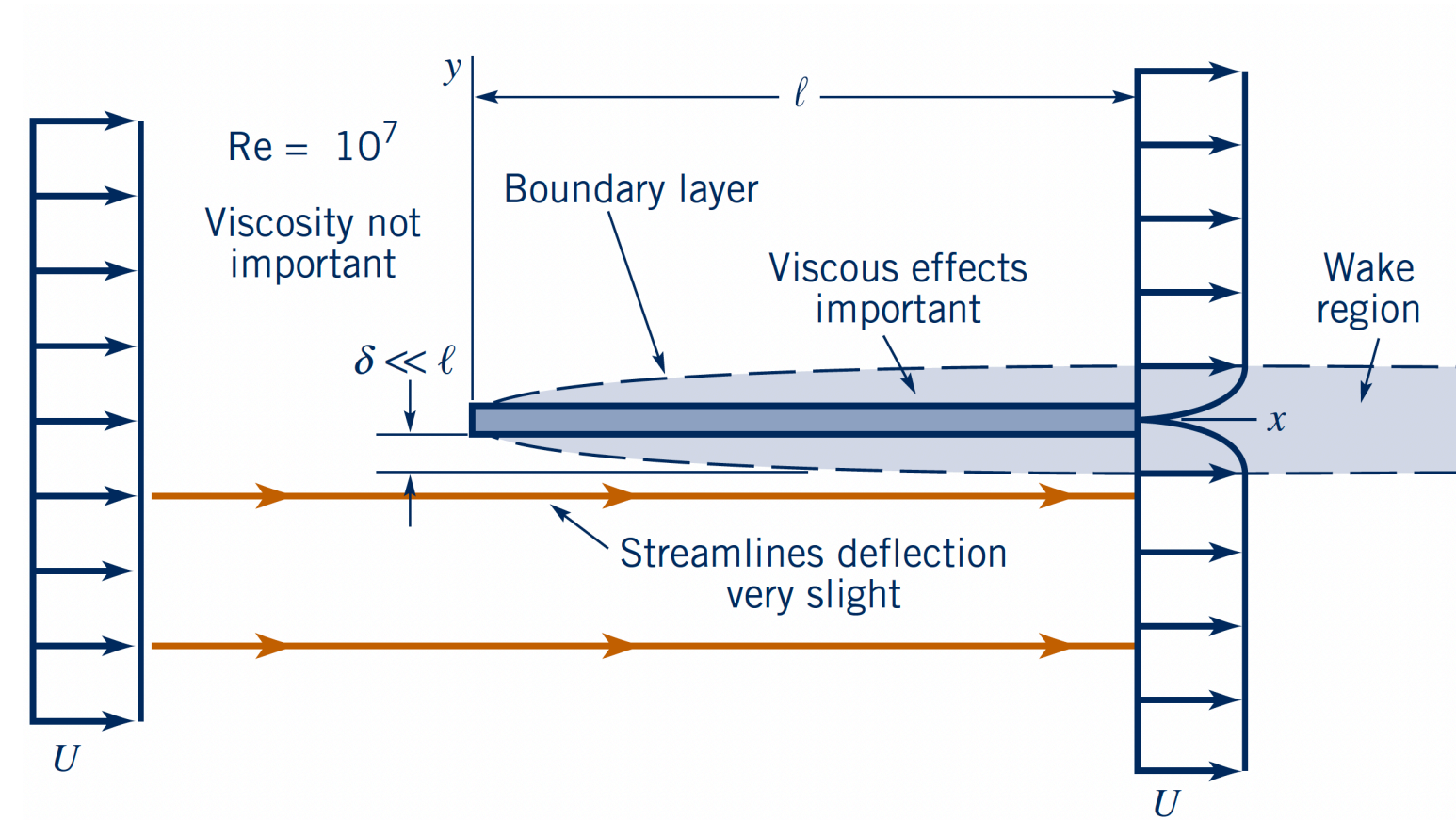
- Plate length: l
- Upstream velocity: U
- Boundary layer: $u < 0.99U$



Re -dependency for streamlined body

Parallel flow to a flat plate at three different $Re = \{0.1, 10, 10^7\}$.

- Plate length: l
- Upstream velocity: U
- Boundary layer: $u < 0.99U$
- Small viscous area
 - Thin BL: $\delta_{99} \ll l$
 - Wake region



Re -dependency for bluff body

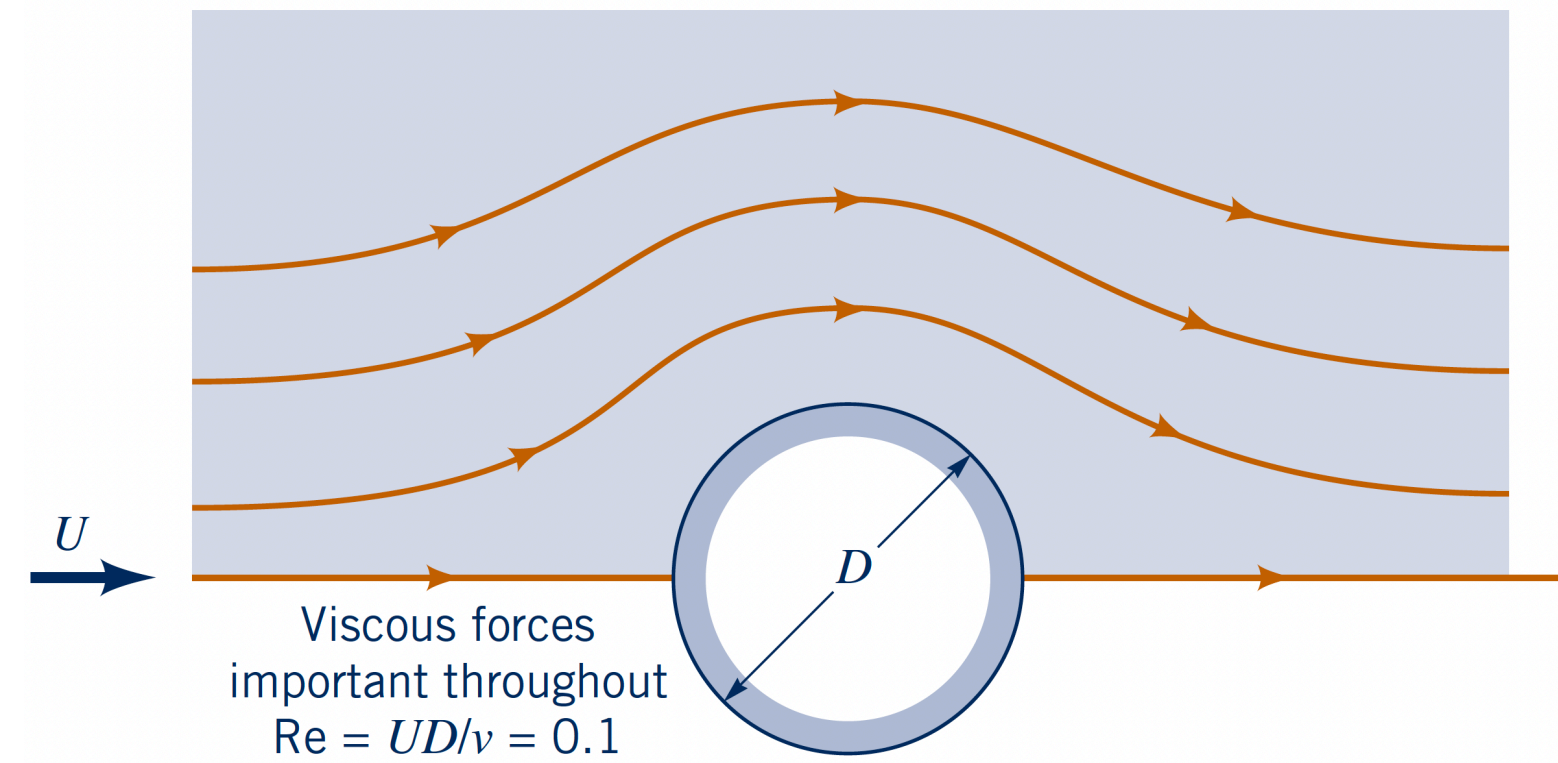
Flow past a circular cylinder at (same) three $Re = \{0.1, 10, 10^7\}$.

- Diameter: D
- Upstream velocity: U
- Boundary layer: $u < 0.99U$

Re -dependency for bluff body

Flow past a circular cylinder at (same) three $Re = \{0.1, 10, 10^7\}$.

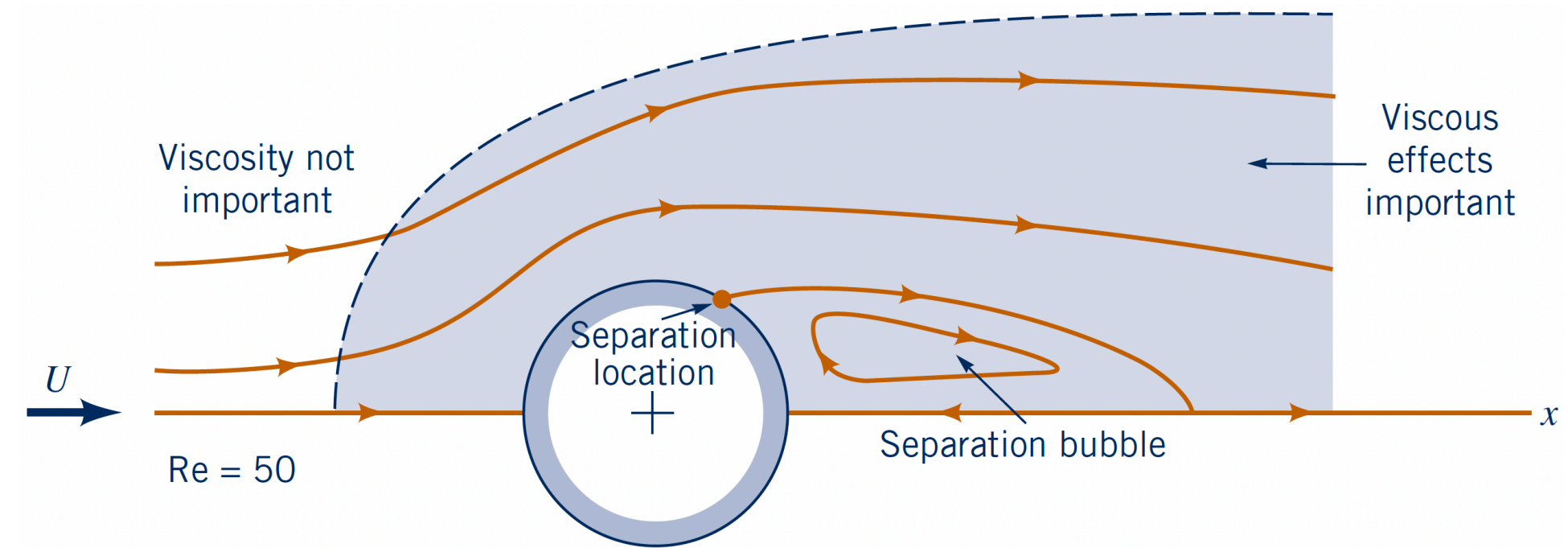
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Re -dependency for bluff body

Flow past a circular cylinder at (same) three $Re = \{0.1, 10, 10^7\}$.

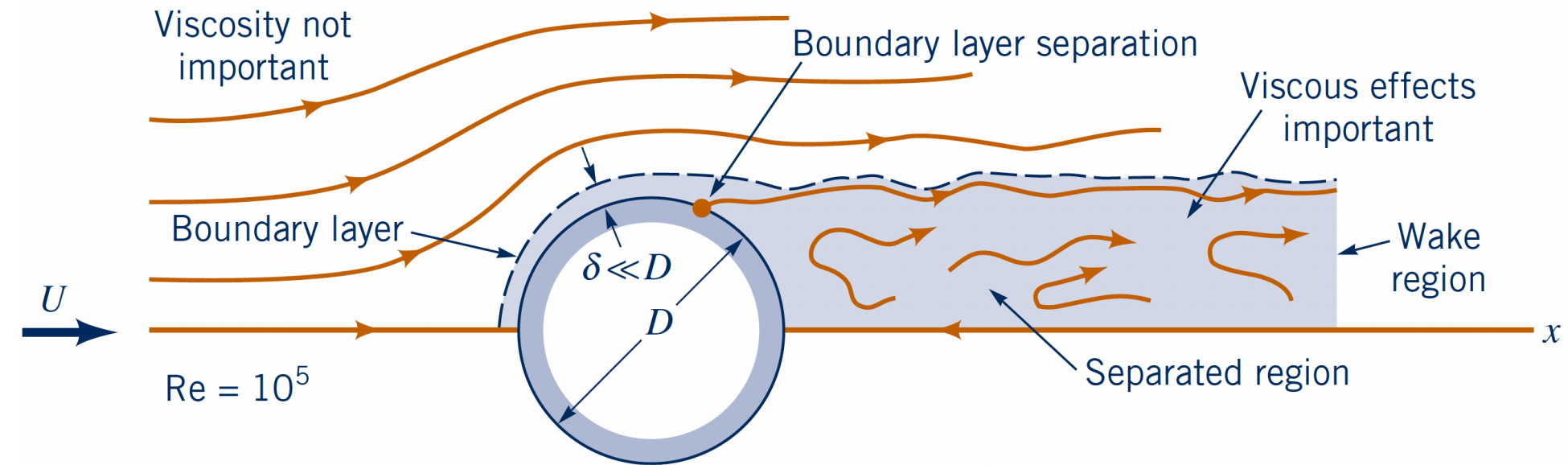
- Diameter: D
- Upstream velocity: U
- Boundary layer: $u < 0.99U$
- *Separation* occurs
 - Inertia too large to follow contour of body
 - Recirculation region appears



Re -dependency for bluff body

Flow past a circular cylinder at (same) three $Re = \{0.1, 10, 10^7\}$.

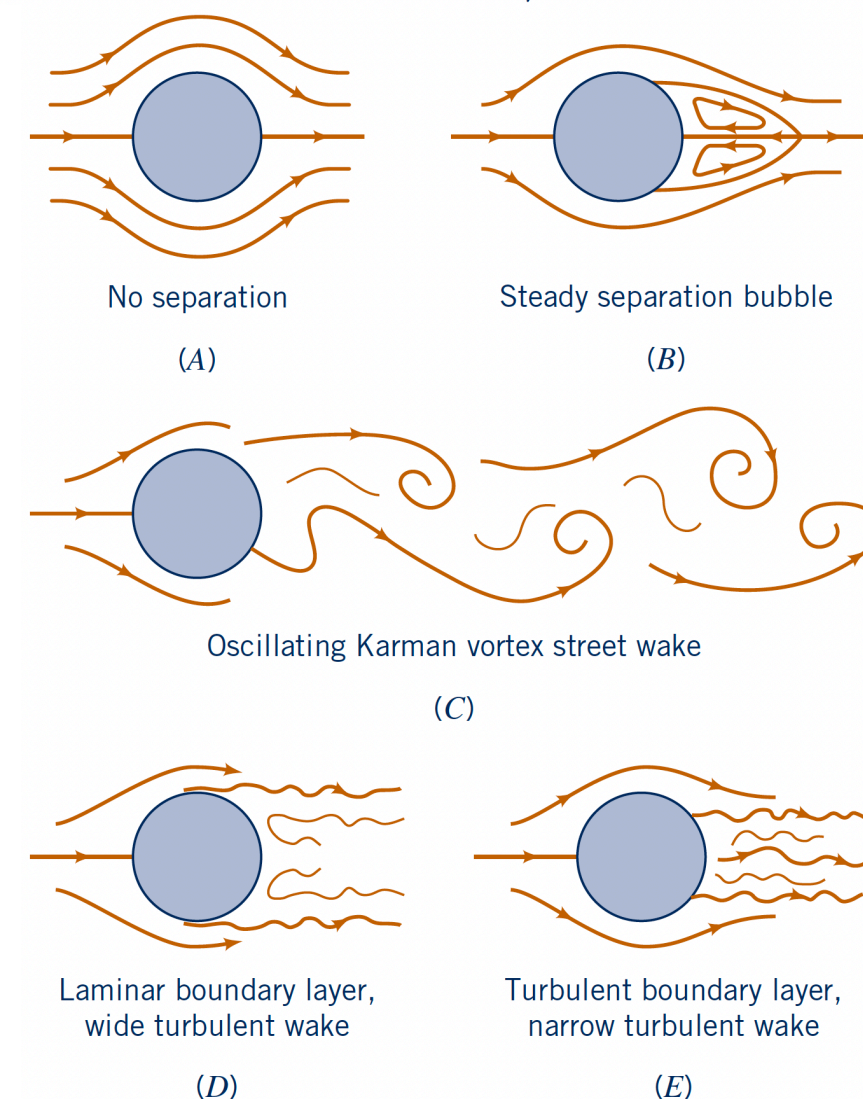
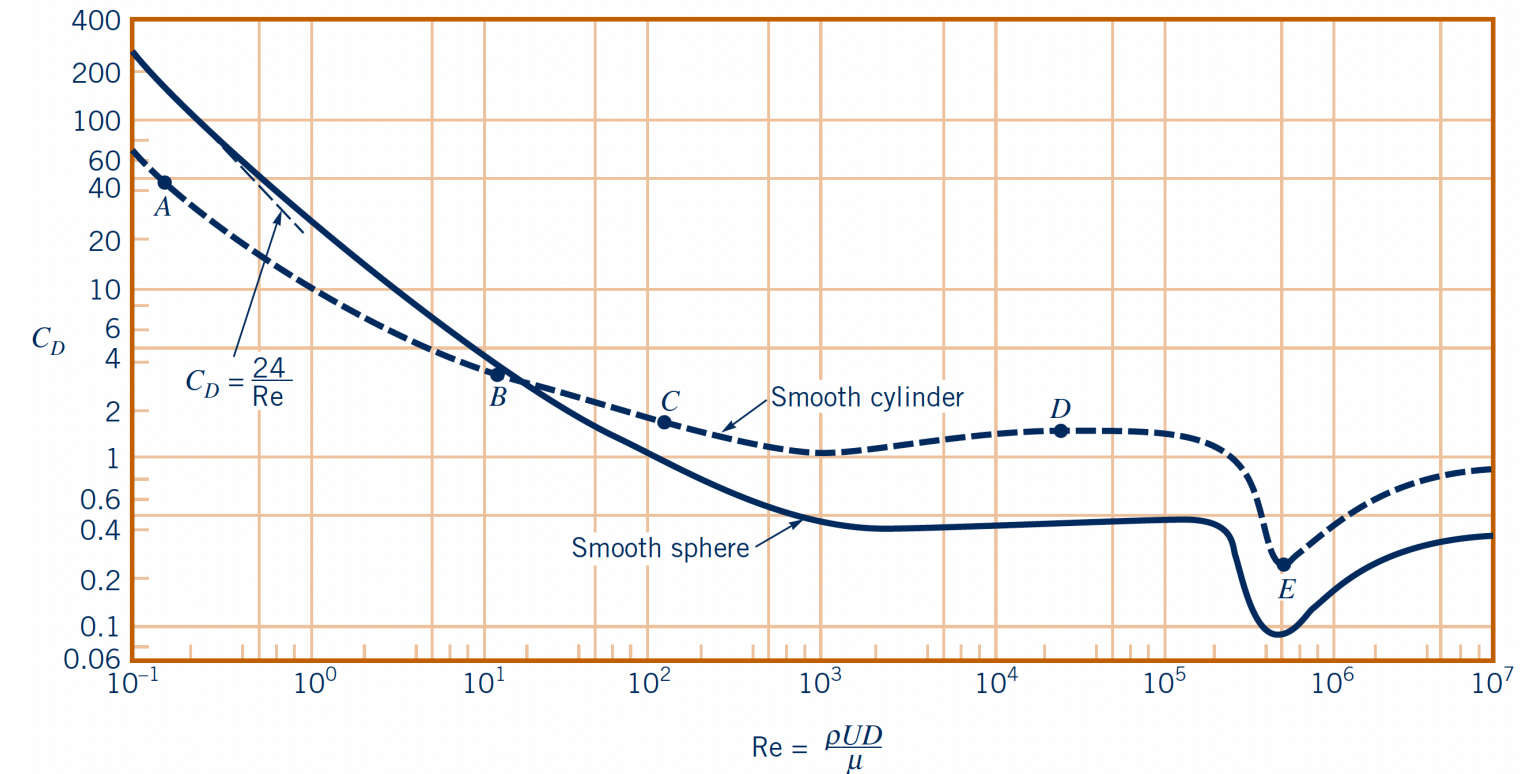
- Diameter: D
- Upstream velocity: U
- Boundary layer: $u < 0.99U$
- *Separation* forms a turbulent wake
 - Thin BL: $\delta_{99} \ll l$
- In a CFD simulation which region of this flow do you think is most difficult to accurately model?
 - The inviscid region, the boundary layer, or the separated/wake region?
 - For more info refer to (e.g.) papers on *Detached Eddy Simulation (DES)*



Re-dependency: Drag Crisis

Closer inspection of the flow past a circular cylinder and sphere as a function of *Re*.

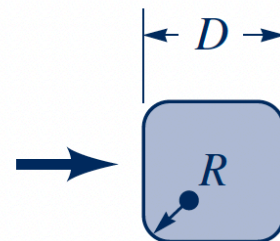
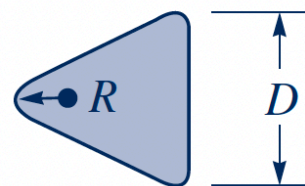
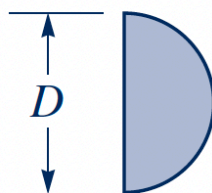
- At very low $Re < 1$: Drag scales with the characteristic *viscous stress* $\mu U/l$, meaning that, $C_D \propto 1/Re$
- At moderate/high $Re > 10^3$: Drag scales with the *dynamic pressure* $\rho U^2/2$ and C_D is largely independent of *Re*
 - Except during the "Drag Crisis" (E): A sudden large drop of C_D at $10^5 < Re < 10^6$
- Surface roughness can trigger turbulent BL at lower *Re*
 - Typical golf balls: $10^4 < Re < 10^5$
 - Why do golf balls have dimples?*



Force coefficients: Where to find them

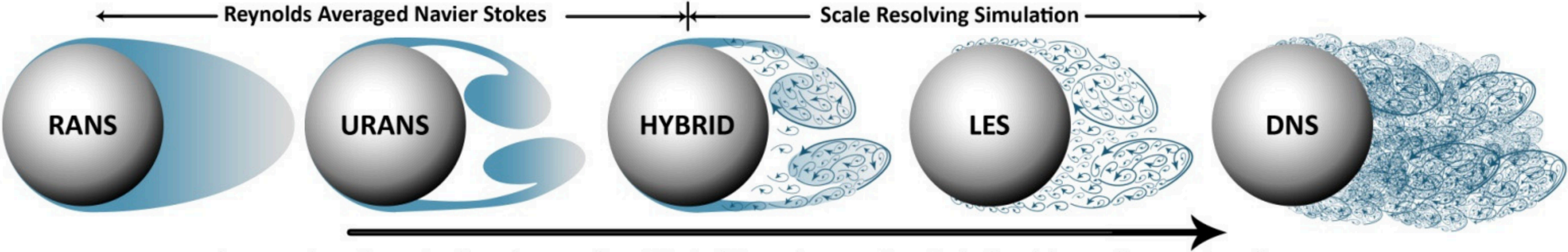
Force coefficients: Drag, lift, added mass coefficients have been determined experimentally or numerically for many typical geometries and flow conditions

- Example: Drag coefficients for 2-D bodies
- Tables with force coefficients can be found in various standards, e.g.
 - [DNV RP C205](#)
- Or fluid mechanics textbooks:
 - [Hoerner \(1965\)](#)
 - [Blevins \(1984\)](#)

Shape	Reference area A (b = length)	Drag coefficient $C_D = \frac{\mathcal{D}}{\frac{1}{2}\rho U^2 A}$	Reynolds number $Re = \rho UD/\mu$																		
 Square rod with rounded corners	$A = bD$	<table><tr><th>R/D</th><th>C_D</th></tr><tr><td>0</td><td>2.2</td></tr><tr><td>0.02</td><td>2.0</td></tr><tr><td>0.17</td><td>1.2</td></tr><tr><td>0.33</td><td>1.0</td></tr></table>	R/D	C_D	0	2.2	0.02	2.0	0.17	1.2	0.33	1.0	$Re = 10^5$								
R/D	C_D																				
0	2.2																				
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 Rounded equilateral triangle	$A = bD$	<table><tr><th>R/D</th><th colspan="2">C_D</th></tr><tr><td></td><th>→</th><th>←</th></tr><tr><td>0</td><td>1.4</td><td>2.1</td></tr><tr><td>0.02</td><td>1.2</td><td>2.0</td></tr><tr><td>0.08</td><td>1.3</td><td>1.9</td></tr><tr><td>0.25</td><td>1.1</td><td>1.3</td></tr></table>	R/D	C_D			→	←	0	1.4	2.1	0.02	1.2	2.0	0.08	1.3	1.9	0.25	1.1	1.3	$Re = 10^5$
R/D	C_D																				
	→	←																			
0	1.4	2.1																			
0.02	1.2	2.0																			
0.08	1.3	1.9																			
0.25	1.1	1.3																			
 Semicircular shell	$A = bD$	<table><tr><td>→</td><td>2.3</td></tr><tr><td>←</td><td>1.1</td></tr></table>	→	2.3	←	1.1	$Re = 2 \times 10^4$														
→	2.3																				
←	1.1																				
 Semicircular cylinder	$A = bD$	<table><tr><td>→</td><td>2.15</td></tr><tr><td>←</td><td>1.15</td></tr></table>	→	2.15	←	1.15	$Re > 10^4$														
→	2.15																				
←	1.15																				

Hoerner (1965) & Blevins (1984)

Approaches to Turbulence Modeling



Hart et al. (2016)